

Northern University of Business and Technology Khulna
Department of Computer Science and Engineering

Feature Completion, Integration & Mid-Project Review

Project Title: AI-Powered Resume Analyzer and Career Copilot

Course Title: Mobile Computing Lab

Course Code: CSE4204

Submitted to:

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Assistant Professor

Department of Computer Science and Engineering
Northern University of Business and Technology, Khulna

TEAM INFORMATION SHEET

Field	Details
Official Team Name	CSE4204-8A-T05
Section	8A
Project Title	AI-Powered Resume Analyzer and Career Copilot
Team Member List with IDs	1. Mahammad Hasan (ID:11220320828) -Team Leader, Frontend Architecture Lead 2. Shanawaz Sakib (ID:11220320916) - Core System Integration Engineer 3. Rasel Ratul (ID:11220320827) - UI-UX & Tailwind Matrix Designer 4. Ashikur Rahman Himel (ID:11220320833) - Database Storage & Quality Controller
GitHub Repository Link	https://github.com/mahamud894/AI_Resume_Analyzer
Figma Link	https://www.figma.com/design/Ls0B3DIjxRKrP2RIzYQxPG/AI-Resume-Analyzer?node-id=2-2&t=IFEVQCoUHt0JSumI-1

1. Feature Completion Checklist & Status Summary

Major Feature / Module	Status	Reason for Incomplete Status / Scope Note	Planned Completion Date
User Authentication (Puter.js OAuth)	Completed	Fully functional passwordless sign-in, session synchronization, and secure route guarding.	Completed
Landing & Navigation UI	Completed	Fully responsive Navbar (Navbar.tsx) featuring dynamic auth-state buttons built with Tailwind CSS v4.	Completed
PDF Resume Upload & Parsing	Completed	Integrated react-dropzone and client-side pdfjs-dist libraries for robust raw text extraction and file validation.	Completed
AI Resume Evaluation (ATS & Skills)	Completed	Seamlessly connected via Puter.js AI gateway (puter.ai.chat) to execute deep semantic evaluations of multi-page resumes	Completed

		against explicit job descriptions.	
Dashboard & History View	In Progress	Visualizing past analysis cards and fetching records dynamically from local storage document caches.	August 20, 2026
Custom Backend & MongoDB Migration	In Progress	Express.js server routes and Mongoose ODM models are written; currently migrating data flows from the client SDK to self-hosted controllers.	August 25, 2026

2. Detailed System Integration Status

2.1 Frontend–Backend Integration Status

- **Current Operational State:** The React 19 frontend communicates dynamically with the system infrastructure. State management via React Router v7 and Zustand ensures smooth data conveyance across core application pages (app/routes/upload.tsx, app/routes/resume.tsx).
- **API Requests & Data Responses:** Form submissions on the upload view successfully bundle candidate PDF binary streams, Target Company names, Job Titles, and raw Job Descriptions into structured asynchronous payloads.

2.2 AI Integration Status

- **Workflow Embedment:** Artificial Intelligence is deeply integrated into the application's core workflow through our prompt orchestration module (`app/lib/puter.ts`). When a user submits a candidate profile, the text data is evaluated against structured system rules.
- **Security & Response Safety:** API credentials and routing proxies are strictly guarded at the cloud boundary. Fallback response blocks and regex-based markdown sanitizers (`(/^\`\`\`json|```$/g)`) are active to handle malformed JSON strings or network timeout lags gracefully without breaking the UI.

2.3 User Authentication Status

- **Workflow Verified:** Registration, sign-in, active session persistence, and secure logout routines are fully operational using Puter's passwordless authentication mechanism.
- **Protected Routes:** Core analytical panels and dashboard views enforce strict session checks through conditional rendering guards embedded within navigation and layout components.

3. System Architecture Evolution: Shifting to a Custom Server Backend

To achieve enterprise-grade data control, enhanced database persistence, and absolute API security, our engineering team is actively executing a complete architectural migration:

- **Transition Scope:** We are transitioning from Puter's direct client-to-cloud SDK infrastructure to a dedicated **Custom Self-Hosted Node.js / Express.js Backend Cluster** integrated with **MongoDB Atlas (NoSQL)** managed via **Mongoose ODM**.
- **Current Migration Status:** The Express server routing endpoints (`/api/auth`, `/api/resume/upload`, `/api/resume/analyze`) and Mongoose document models (`User.js`, `Analysis.js`) have been written and are undergoing local staging validation.
- **Target Timeline:** The complete transition from the client-side SDK model to the fully self-hosted Express and MongoDB server environment is on track to be finalized within the next **1 to 2 weeks**.
- **Architecture Agnosticism:** The frontend is structured with environment configurations, ensuring that switching from the serverless client mode to the self-hosted server cluster

requires zero refactoring of the user interface components.

4. Major Problems Encountered & Solutions Implemented

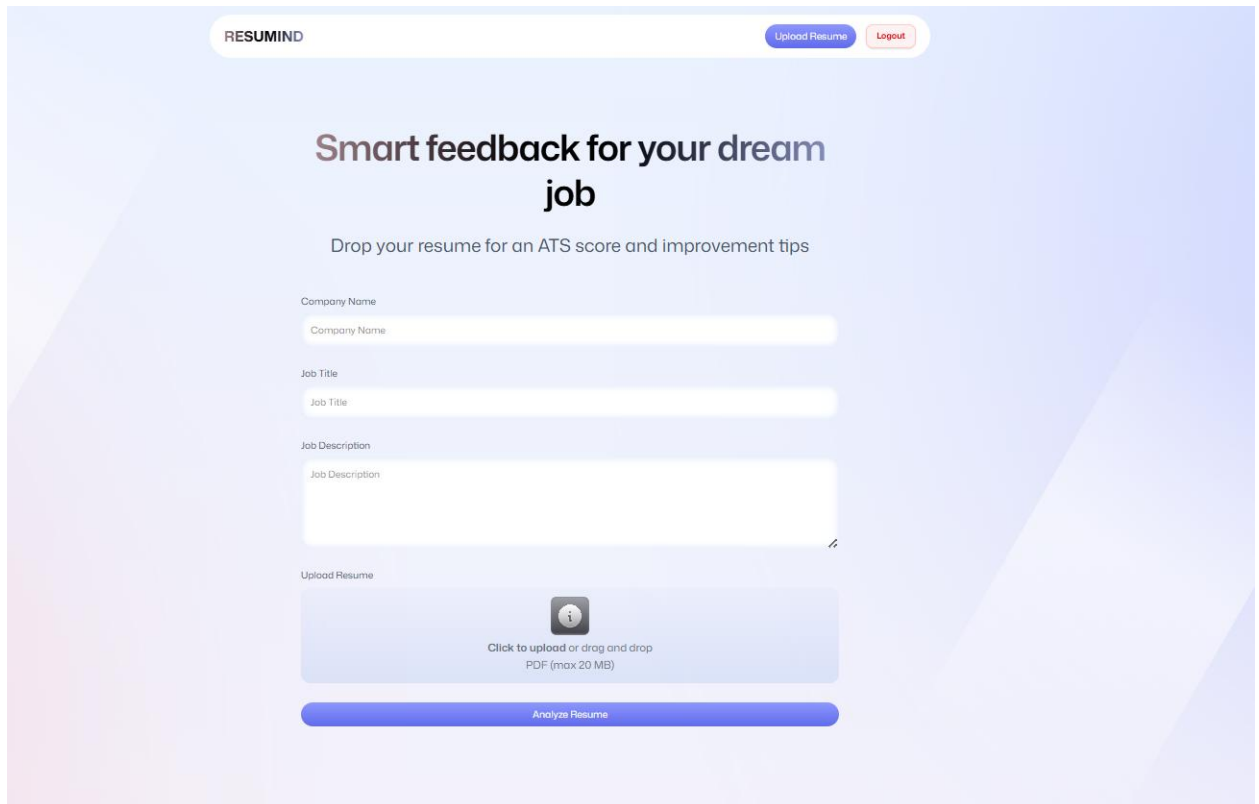
1. **Problem:** Client-side relative import path mismatches during modular component restructuring and file reorganization.
 - **Solution:** Standardized path conventions and utilized Vite TypeScript path aliases (~/) to ensure clean, error-free module resolution.
2. **Problem:** Inconsistent or wrapped JSON formatting returned by large language models occasionally triggering client-side rendering exceptions.
 - **Solution:** Implemented strict regex-based markdown sanitization and defensive try-catch JSON parsing blocks equipped with default fallback schemas.
3. **Problem:** Balancing production stability while migrating the system's underlying data pipeline to a self-hosted Express and MongoDB server architecture.
 - **Solution:** Adopted a hybrid migration strategy—maintaining the active production-ready client workflow while staging the Node.js/Mongoose routes on a separate branch for upcoming deployment.

5. Changes from Original Proposal / Scope Adjustments

- **Scope Modification:** Initially, local browser storage was planned for document caching and state management.
- **Reason for Change:** To improve system portability, security, and enterprise scalability, we integrated cloud file system management (puter.fs) and are concurrently scaling toward MongoDB Atlas for permanent document logging and historical tracking.
- **New Approach:** A hybrid cloud-backed architecture ensuring rapid feature delivery today while paving the way for a fully independent, self-hosted server backend tomorrow.

6. Current Project Screenshots Reference

6.1 Upload Workspace View



The screenshot displays the 'RESUMIND' application interface for uploading a resume. At the top, a navigation bar contains the 'RESUMIND' logo on the left and two buttons, 'Upload Resume' and 'Logout', on the right. The main heading reads 'Smart feedback for your dream job'. Below this, a subheading states 'Drop your resume for an ATS score and improvement tips'. The form includes three input fields: 'Company Name', 'Job Title', and 'Job Description'. The 'Job Description' field is a larger text area with a small icon at the bottom right. Below these fields is an 'Upload Resume' section featuring a large light blue box with a circular icon containing an 'i' and the text 'Click to upload or drag and drop PDF (max 20 MB)'. At the bottom of the form is a prominent blue button labeled 'Analyze Resume'.

Fig: Upload Workspace View



MAHAMUD HASSAN

STUDENT

About Me

Computer Science & Engineering student | Passionate about AI, web development, and cybersecurity.

CONTACT

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KEY SKILLS

- **Front-end:** HTML, CSS, JavaScript
- **Back-end:** PHP, Laravel Framework, Database: MySQL
- **Other Languages:** C++, Java, Python
- **Tools:** Version Control(Git), GitHub, Visual Studio Code, CodeBlocks, Xampp
- **Also:** SEO, Active Writing, Content Writing & Editing, On-Page SEO & Off-Page SEO, Keyword Research & Competitor Analysis, Technical SEO & Link Building, Google Analytics & Search Console, Product Listing & Keyword Research

LANGUAGE

- English
- Bangla

EDUCATION

B.Sc. in Computer Science and Engineering
Northern University of Business and Technology
(2023 to Present (Ongoing))
Expected Graduation: August 2025

H.S.C. :Science | GPA: 4.67
Abdurrahman College, Khulna, Bangladesh
(2019 to 2021)

PROFESSIONAL EXPERIENCE

Professional Training on PHP Laravel Framework | November 30, 2024 to Present
Following Digital Communication and Economy (DCE) program under Bangladesh Computer Council (BCC) under the ICT Division, Government of Bangladesh in HTML, CSS, JavaScript, MySQL, PHP, Laravel Framework.
Project: Course Management System

Final Touch, Bangladesh
SEO Content Writer, Dec 2024 - Present
Developed and implemented SEO strategies that increased organic traffic by 15%.
Conducted keyword research, competitor analysis, and on-page optimization.
Created high-quality Article, SEO friendly content for blogs, websites.

PERSONAL INFORMATION

- **Name:** Mahamud Hassan
- **Father's Name:** Tahirul Mujafer Rahman
- **Mother's Name:** Tasfia Farhin
- **Present Address:** Sher, Baranagar, Khulna, Bangladesh
- **Date of Birth:** 13-11-2003
- **Nationality:** Bangladeshi
- **Religion:** Islam
- **Marital Status:** Unmarried



Your Resume Score

This score is calculated based on the variables listed below.

Tone & Style

Strong

85/100

Content

Strong

75/100

Structure

Strong

90/100

Skills

Strong

80/100

ATS Score - 80/100

Great Job!

This score represents how well your resume is likely to perform in Applicant Tracking Systems used by employers.

👉 Strong mention of tools like Google Analytics and Search Console.

👉 Incorporate more keywords from the job description such as 'monitor rankings' and 'technical SEO optimization' explicitly.

Keep refining your resume to improve your chances of getting past ATS filters and into the hands of recruiters.

Tone & Style

85/100

The screenshot displays the 'AI Resume Analyzer' application interface. The top navigation bar includes the user profile 'msharadk04' and the application name 'AI Resume Analyzer'. Below this, a secondary navigation bar shows tabs for 'main', '3 branches', and '0 tags'. The main content area is titled 'main' and contains a list of files and folders. Each entry includes a file icon, the file name, a brief description, and a timestamp. The sidebar on the right contains several sections: 'About' (no descriptions yet), 'Features' (listing features like Activity, Files, and Everything), 'Reviews' (no reviews yet), 'Packages' (listing packages like react-router-dom and vite), 'Contributors' (listing contributors like msharadk04 and lukaszkuk), 'Languages' (showing a bar chart for JavaScript, TypeScript, and CSS), and 'Suggestive workflows' (listing workflows like 'Create new project' and 'Add new feature').

7. Conclusion & Mid-Review Readiness

By Week 09, our project has successfully moved past isolated module creation. The user authentication, frontend navigation, document parsing, and AI evaluation layers are fully integrated into a unified software system. Furthermore, our active migration toward a self-hosted Node.js/Express and MongoDB server architecture demonstrates high technical maturity. The team is fully prepared for the upcoming mid-project review and individual vivas.